



National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEecs)
Department of Computing (DOC)

LEARN | EXPLORE | SOLVE | CREATE

Mathematical Methods for Computing

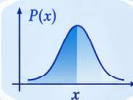
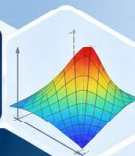
CS-813 | Fall 2026

MSCS Program
SEecs, NUST

Prof. Dr. Quanita Kiran

*Mathematics
Drives
Computing
Forward*

$$Ax = b$$
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\min_{\theta} f(\theta)$$



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A BRIGHTER TOMORROW

VECTORS AND SCALARS

Mathematics
— for a Smarter
Tomorrow

1 Scalar

- A **scalar** is an individual numerical value, typically drawn from the real numbers ($\mathbb{R}$).
- It often represents the value of a **single attribute** such as age, temperature, salary, etc.
- Scalars are written using lowercase letters such as a , b , x .
- Examples:** 5, -3.2 , 0, 25, 100.

2 Vector

- A **vector** represents a given set of data as an ordered array of numerical values (scalars).
- The individual values in the array are called the **components** of the vector.
- The number of components is called the vector **dimension** (or length).
- Vectors are usually written using bold letters ($\mathbf{x}$) or as an array: $\mathbf{x} = [x_1, x_2, \dots, x_n]$.

Examples of Scalars (Attributes)

Age	Hourly Wage	Years of Experience	Temperature	Height (m)
25	30	5	36.5	1.75

Example: A 3-Dimensional Vector

A 3-dimensional vector representation of a 25-year-old person making 30 dollars an hour, and having 5 years of experience:

$$\mathbf{x} = [25, 30, 5]$$

$x_1 = 25$ (Age)
 $x_2 = 30$ (Hourly Wage)
 $x_3 = 5$ (Years of Experience)
Dimension: 3

More Examples

2-dimensional vector
(representing a point in 2D):

$$\mathbf{x} = [3, -1]$$

Dimension: 2

4-dimensional vector
(e.g., features of a house):

$$\mathbf{x} = [1200, 3, 2, 250000]$$

Dimension: 4

n -dimensional vector
(general form):

$$\mathbf{x} = [x_1, x_2, \dots, x_n]$$

Dimension: n

Scalars are individual numbers, while vectors are ordered collections of numbers.



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Mathematical Methods for Computing (CS-813)

Vector Addition

Combining Vectors to Form New Vectors

Mathematics
Builds a Stronger
Tomorrow

1 Definition

The sum of two vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$ is a vector obtained by adding their corresponding components:

$$\mathbf{x} + \mathbf{y} = [x_1 + y_1, x_2 + y_2, \dots, x_d + y_d]$$

- The vectors $\mathbf{x}$ and $\mathbf{y}$ must have the same dimension.
- Vector addition is **commutative**: $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$.
- Vector addition is **associative**:
 $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$.
- The zero vector $\mathbf{0} = [0, 0, \dots, 0]$ satisfies
 $\mathbf{x} + \mathbf{0} = \mathbf{x}$.

2 Numerical Example

Example 1 (3-dimensional vectors)

Let $\mathbf{x} = [2, -1, 3]$ and $\mathbf{y} = [4, 5, -2]$. Then,

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= [2 + 4, -1 + 5, 3 + (-2)] \\ &= [6, 4, 1].\end{aligned}$$

Result:

$$\mathbf{x} + \mathbf{y} = [6, 4, 1]$$

Example 2 (2-dimensional vectors)

Let $\mathbf{u} = [3, 2]$ and $\mathbf{v} = [-1, 4]$. Then,

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= [3 + (-1), 2 + 4] \\ &= [2, 6].\end{aligned}$$

Result:

$$\mathbf{u} + \mathbf{v} = [2, 6]$$

Note: Vector addition combines the corresponding components of the vectors to produce a new vector with the same dimension.



Geometric Interpretation of Vector Addition (2D)

Adding Vectors as a Parallelogram

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Vectors in 2D

A 2D vector is represented as

$$\mathbf{x} = (x_1, x_2)$$

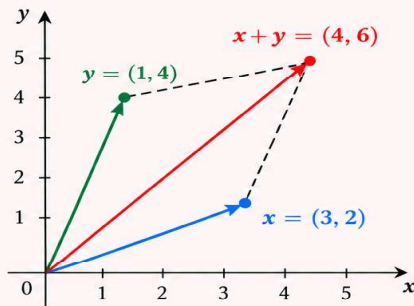
Geometrically, it is an arrow from the origin to the point (x_1, x_2) in the plane.

Key Observations

- 1 The vector $\mathbf{x} + \mathbf{y}$ is the diagonal of the parallelogram with $\mathbf{x}$ and $\mathbf{y}$ as adjacent sides.
- 2 The order of addition does not change the result:
 $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$.
- 3 Both the parallelogram rule and the tip-to-tail rule give the same resultant vector.

Vector Addition (Parallelogram Rule)

Let $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (1, 4)$. Their sum $\mathbf{x} + \mathbf{y}$ is the diagonal of the parallelogram formed by $\mathbf{x}$ and $\mathbf{y}$.



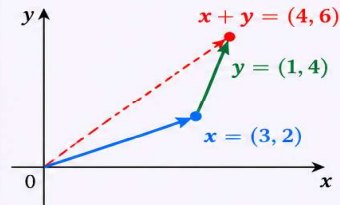
Component-wise Addition

$$\mathbf{x} = (3, 2), \quad \mathbf{y} = (1, 4)$$

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (3 + 1, 2 + 4) \\ &= (4, 6)\end{aligned}$$

Tip-to-Tail Rule (Equivalent View)

Place $\mathbf{y}$ at the head of $\mathbf{x}$.



Vector addition combines both magnitude and direction to produce a new vector.

Graphical representation helps us visualize vector operations in the plane.

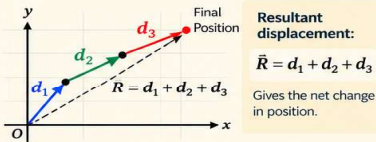
Applications of Vector Addition

Combining Quantities in the Real World

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1 Displacement and Navigation

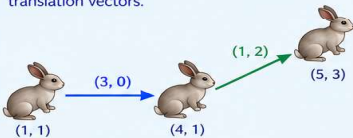
Find the final position after multiple moves by adding displacement vectors.



Used in navigation, robotics and physics.

4 Computer Graphics and Animation

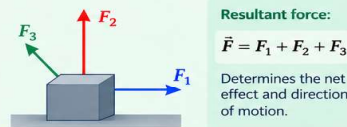
Object movements are combined using translation vectors.



Used for object transformations, animations and game development.

2 Forces in Engineering

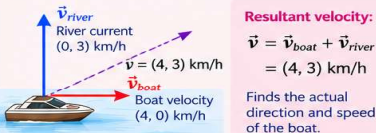
The total force on an object is the vector sum of individual forces.



Used in mechanics, structural analysis and dynamics.

3 Velocity and Motion

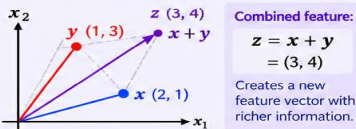
The final velocity is the vector sum of individual velocity components.



Used in transportation, aviation and oceanography.

5 Data Analysis and Machine Learning

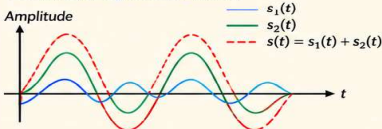
Feature vectors can be combined to create new representations (e.g., document embeddings).



Used in clustering, dimensionality reduction, recommendation systems and similarity analysis.

6 Signal Processing and Communications

Signals (represented as vectors) can be combined using vector addition.



Used in signal synthesis, noise reduction and wireless communications.



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VECTOR SUBTRACTION

1 Definition

The difference of two vectors

$$\mathbf{x} = [x_1, x_2, \dots, x_d] \quad \text{and} \quad \mathbf{y} = [y_1, y_2, \dots, y_d]$$

is the vector obtained by subtracting their corresponding components:

$$\mathbf{x} - \mathbf{y} = [x_1 - y_1, x_2 - y_2, \dots, x_d - y_d]$$

- The vectors $\mathbf{x}$ and $\mathbf{y}$ must have the same dimension.
- Vector subtraction is not commutative: $\mathbf{x} - \mathbf{y} \neq \mathbf{y} - \mathbf{x}$.
- It can be written as $\mathbf{x} - \mathbf{y} = \mathbf{x} + (-\mathbf{y})$.
- The zero vector $\mathbf{0} = [0, 0, \dots, 0]$ satisfies $\mathbf{x} - \mathbf{x} = \mathbf{0}$.

Vector subtraction gives the displacement from one vector to another.

2 Numerical Example

$$\text{Let } \mathbf{x} = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}$$

$$\text{Then, } \mathbf{x} - \mathbf{y} = \begin{bmatrix} 4 - 1 \\ 3 - 5 \\ -2 - 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}$$

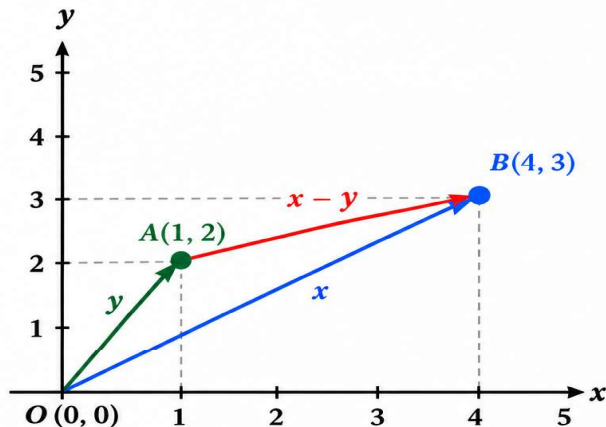
Therefore,

$$\mathbf{x} - \mathbf{y} = \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}.$$



Geometric Interpretation of Vector Subtraction (2D)

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Example

Let $x = (4, 3)$ and $y = (1, 2)$.

Then,

$$x - y = (4 - 1, 3 - 2) = (3, 1).$$

Geometric Interpretation

- The vector $x - y$ goes from the tip of y (point A) to the tip of x (point B).
- It represents the displacement needed to move from y to x .
- In this example, $x - y = (3, 1)$, which means move 3 units in the positive x -direction and 1 unit in the positive y -direction.



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Applications of Vector Subtraction

Finding Differences, Displacements and Relative Information

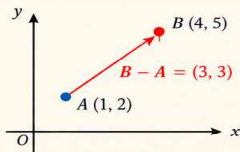
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1 Physics – Displacement and Relative Motion

The displacement from point A to point B is given by the vector difference:

$$\vec{AB} = B - A$$

It gives the change in position, distance and direction.



Used to compute displacement, velocity change and relative position in mechanics and navigation.

3 Machine Learning – Comparing Data Points

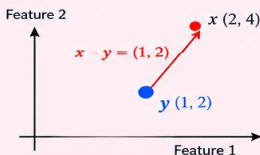
The difference between two feature vectors measures how far apart two data points are in the feature space.

$$d = x - y$$

Example:

$$x = [2, 4, 1], \quad y = [1, 2, 3]$$

$$x - y = [1, 2, -2]$$



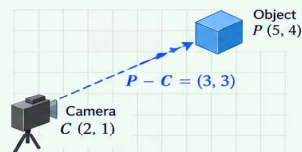
Used in distance calculations, nearest neighbor search, clustering and similarity measures.

2 Computer Graphics – Relative Position and Motion

The relative position of an object with respect to a reference point is found using vector subtraction.

$$\text{relative position} = P - C$$

Useful for camera movement, object transformations, and collision detection.

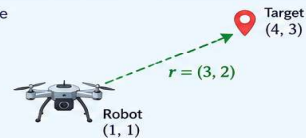


Vector subtraction helps determine relative locations and directions in 2D/3D graphics and animations.

4 Engineering & Robotics – Relative Positioning

Vector subtraction gives the relative position between two objects, which is essential for guidance, control and path planning.

$$r = p_{\text{target}} - p_{\text{robot}}$$



Used in robotics, autonomous navigation, and control systems to reach targets or avoid obstacles.



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SCALAR MULTIPLICATION OF A VECTOR

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1 Definition

For a vector $\mathbf{x} = [x_1, x_2, \dots, x_d] \in \mathbb{R}^d$ and a scalar $k \in \mathbb{R}$, the scalar multiplication of $\mathbf{x}$ by k is defined as

$$k\mathbf{x} = [kx_1, kx_2, \dots, kx_d].$$

- Each component of the vector is multiplied by the scalar k .
- The vector $k\mathbf{x}$ is always collinear with $\mathbf{x}$ (lies on the same line through the origin).
- If $k > 0$, the direction remains the same.
- If $k < 0$, the direction is reversed.
- If $k = 0$, the zero vector is obtained: $\mathbf{0x} = \mathbf{0}$.

2 Numerical Example

Let $\mathbf{x} = [2, -1, 3]$. Find $k\mathbf{x}$ for the following values of k .

- For $k = 2$:
$$2\mathbf{x} = 2[2, -1, 3] = [4, -2, 6].$$
- For $k = \frac{1}{2}$:
$$\frac{1}{2}\mathbf{x} = \frac{1}{2}[2, -1, 3] = \left[1, -\frac{1}{2}, \frac{3}{2}\right].$$
- For $k = -1$:
$$-\mathbf{x} = -1[2, -1, 3] = [-2, 1, -3].$$
- For $k = 0$:
$$\mathbf{0x} = 0[2, -1, 3] = [0, 0, 0].$$

All these vectors are scalar multiples of $\mathbf{x}$ and lie on the same line through the origin.



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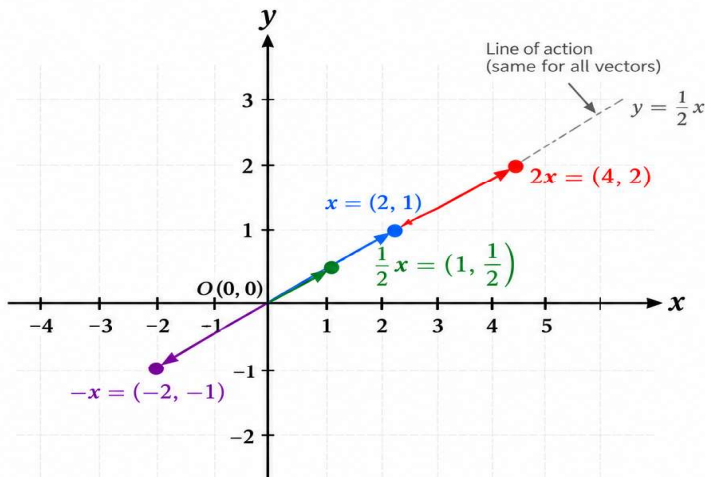


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Mathematical Methods for Computing (CS-813)

Geometric Representation of Scalar Multiplication (2D)

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Example

Let $\mathbf{x} = (2, 1)$. Then

- For $k = 2$: $2\mathbf{x} = (4, 2)$
- For $k = \frac{1}{2}$: $\frac{1}{2}\mathbf{x} = (1, \frac{1}{2})$
- For $k = -1$: $-\mathbf{x} = (-2, -1)$
- For $k = 0$: $0\mathbf{x} = (0, 0)$

Geometric Interpretation

- All the vectors $\mathbf{x}$, $2\mathbf{x}$, $\frac{1}{2}\mathbf{x}$ and $-\mathbf{x}$ lie on the same line of action (the line through the origin in the direction of $\mathbf{x}$).
- If $k > 0$, the vector is in the same direction and its length is scaled by k .
- If $0 < k < 1$, the vector is shorter but in the same direction.
- If $k < 0$, the vector is in the opposite direction but still on the same line.
- If $k = 0$, the zero vector is at the origin.

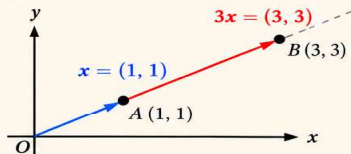
Scalar multiplication changes only the magnitude (length) of a vector, but not its line of action (direction).

Applications of Scalar Multiplication of a Vector

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1 Displacement in Physics

Scalar multiplication scales a displacement vector, keeping the same direction.

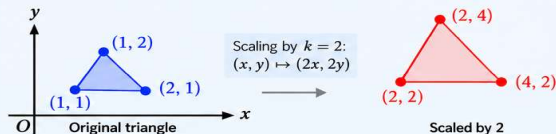


The vector $3\mathbf{x}$ represents a displacement three times the original, in the same direction.

Used to model motion, position and distance (e.g., velocity, acceleration).

3 Scaling and Transformations in Computer Graphics

Scalar multiplication is used to scale objects in 2D/3D graphics by changing their size while preserving their shape and orientation.



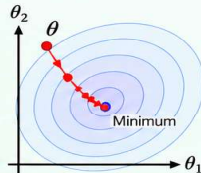
Used for object scaling, zooming and geometric transformations in computer graphics.

2 Gradient Descent in Machine Learning

In gradient descent, we move the parameters in the opposite direction of the gradient, scaled by a learning rate $\eta > 0$.

$$\theta_{new} = \theta - \eta \nabla L(\theta)$$

- The gradient $\nabla L(\theta)$ indicates the direction of steepest increase of the loss function.
- Multiplying by the scalar η controls the step size (how far we move).
- A smaller η takes shorter steps; a larger η takes bigger steps in the same direction.



Used for training machine learning models (e.g., linear regression, neural networks).

- Multiplying position vectors by a scalar changes the size of objects.
- Used for zooming, resizing and modeling transformations in 2D/3D graphics and game development.



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SCALAR (DOT) PRODUCT OF VECTORS

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for Computing

1 Definition

For two vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$ in $\mathbb{R}^d$, the scalar (or dot) product is a real number defined as

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^d x_i y_i = \mathbf{x}^T \mathbf{y}.$$

- The dot product takes two vectors and returns a scalar (a real number).
- It is commutative: $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- It is distributive: $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$.
- $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2 \geq 0$, and $\mathbf{x} \cdot \mathbf{x} = 0$ only if $\mathbf{x} = \mathbf{0}$.

Interpretation:

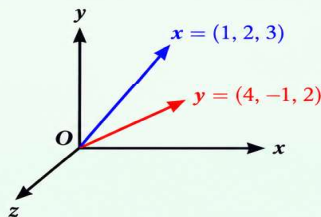
The dot product measures how much two vectors point in the same direction.

2 Numerical Example

Let $\mathbf{x} = [1, 2, 3]$ and $\mathbf{y} = [4, -1, 2]$ in $\mathbb{R}^3$.
Then

$$\begin{aligned}\mathbf{x} \cdot \mathbf{y} &= 1(4) + 2(-1) + 3(2) \\ &= 4 - 2 + 6 \\ &= 8.\end{aligned}$$

Result: $\mathbf{x} \cdot \mathbf{y} = 8$ (a scalar).



Note:

A larger dot product indicates that the vectors are more aligned (smaller angle between them).

Geometric Interpretation of the Dot Product in 2D

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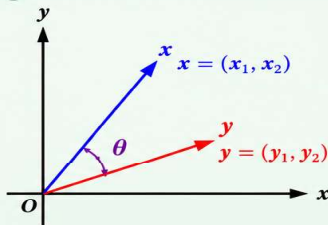
1 Setting

Let $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ be two vectors in $\mathbb{R}^2$. Let θ be the angle between them ($0^\circ \leq \theta \leq 180^\circ$).

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta.$$

- $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ are the lengths (magnitudes) of the vectors.
- θ is the angle between the vectors.
- The dot product measures how much the two vectors point in the same direction.

2 Vectors and Angle in 2D



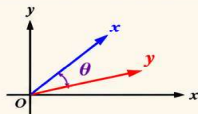
3 What the Formula Means

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$$

- The dot product is the product of the magnitudes of the vectors and the cosine of the angle between them.
- It is large and positive when the vectors are closely aligned (small angle).
- It is zero when the vectors are perpendicular ($\theta = 90^\circ$).
- It is negative when the vectors point in opposite directions (angle greater than 90°).

4 Special Cases (Geometric View)

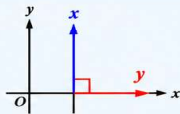
(a) Acute Angle ($0^\circ < \theta < 90^\circ$)



$$\cos \theta > 0 \Rightarrow \mathbf{x} \cdot \mathbf{y} > 0$$

Vectors point in a similar direction.

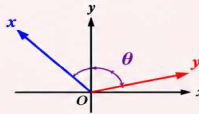
(b) Orthogonal ($\theta = 90^\circ$)



$$\cos \theta = 0 \Rightarrow \mathbf{x} \cdot \mathbf{y} = 0$$

Vectors are perpendicular.

(c) Obtuse Angle ($90^\circ < \theta < 180^\circ$)



$$\cos \theta < 0 \Rightarrow \mathbf{x} \cdot \mathbf{y} < 0$$

Vectors point in opposite directions (to some extent).

5 Example in 2D

Let $\mathbf{x} = (2, 1)$ and $\mathbf{y} = (4, 3)$. Compute the dot product and the angle.

Solution:

- $\mathbf{x} \cdot \mathbf{y} = 2(4) + 1(3) = 11$.
- $\|\mathbf{x}\| = \sqrt{2^2 + 1^2} = \sqrt{5}$, $\|\mathbf{y}\| = \sqrt{4^2 + 3^2} = 5$.
- $\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{11}{\sqrt{5} \cdot 5} = \frac{11}{5\sqrt{5}} \approx 0.983$.
- $\theta = \cos^{-1}(0.983) \approx 10.4^\circ$.

Result: $\mathbf{x} \cdot \mathbf{y} = 11$ and the angle $\theta \approx 10.4^\circ$ (the vectors are closely aligned).

Applications of the Dot Product in Computing and Machine Learning

Mathematical Methods
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1 Measuring Similarity

The dot product measures how similar two vectors are. It is the basis of cosine similarity.

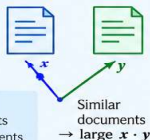
$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Used in:

- Text similarity (NLP)
- Recommendation systems
- Image and document retrieval

Example: Document vectors

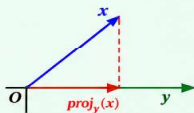
$\cos \theta$ close to 1 $\rightarrow$ similar documents
 $\cos \theta$ close to 0 $\rightarrow$ unrelated documents



2 Projection

The dot product helps compute the projection of one vector onto another.

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{y}\|^2} \mathbf{y}$$



Used in:

- Least squares regression
- Dimensionality reduction (e.g., PCA)
- Feature extraction

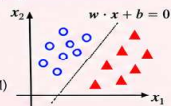
3 Linear Models and Classification

Many ML models compute a linear combination using the dot product.

$$\hat{\mathbf{y}} = \mathbf{w} \cdot \mathbf{x} + b = \sum_{i=1}^d w_i x_i + b$$

Used in:

- Linear regression
- Logistic regression
- Support Vector Machines (SVM)
- Neural network layers (fully connected)



The decision boundary is defined using a dot product.

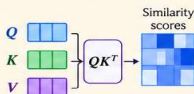
4 Attention Mechanisms in Deep Learning

In Transformers, the dot product computes the similarity between query and key vectors.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Used in:

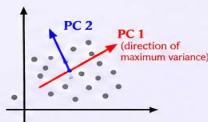
- Natural Language Processing
- Vision Transformers
- Sequence modeling



5 Dimensionality Reduction (PCA)

PCA uses dot products to compute covariance and to project data onto principal components.

$$\text{Cov}(X) = \frac{1}{n} X^T X$$



Used in:

- PCA
- Latent Semantic Analysis (LSA)
- Data compression
- Visualization

6 Other Applications



Graphics and Computer Vision

Compute angles, lighting (normals), and shading using dot products.



Information Retrieval

Search and ranking using vector similarity.



Neural Networks

Dot products in each layer for computing weighted sums.



Physics and Simulation

Compute work: $W = \mathbf{F} \cdot \mathbf{d}$
Measure alignment of forces and movement.



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Mathematical Methods for Computing (CS-813)

Euclidean Norm (Length of a Vector)

Mathematics
for a Smarter Tomorrow

Consider a d -dimensional vector $\mathbf{x} = [x_1, x_2, \dots, x_d] \in \mathbb{R}^d$.

1 Definition

The dot product of a vector $\mathbf{x}$ with itself is called its squared norm (or Euclidean norm). It defines the vector length and is denoted by $\|\mathbf{x}\|$.

$$\|\mathbf{x}\|^2 = \sum_{i=1}^d x_i^2 = \mathbf{x} \cdot \mathbf{x}$$

- The Euclidean norm is always non-negative: $\|\mathbf{x}\| \geq 0$.
- $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = 0$.

2 Geometric Interpretation

The norm of a vector is the Euclidean distance of its coordinates from the origin.

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}$$

3 Normalization (Unit Vector)

Often, vectors are normalized to unit length by dividing them with their norm:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|} = \frac{\mathbf{x}}{\sqrt{\mathbf{x} \cdot \mathbf{x}}}$$

- The vector $\hat{\mathbf{x}}$ has unit length: $\|\hat{\mathbf{x}}\| = 1$.
- Normalization keeps the direction the same, only the magnitude changes to 1.

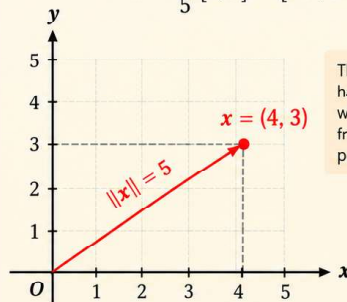
4 Numerical Example (2D)

Let $\mathbf{x} = [4, 3]$. Then

$$\|\mathbf{x}\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.$$

The unit vector is

$$\hat{\mathbf{x}} = \frac{1}{5} [4, 3] = [0.8, 0.6].$$



The vector $\mathbf{x} = (4, 3)$ has Euclidean norm 5, which is the distance from the origin to the point $(4, 3)$.



Euclidean Distance Between Two Vectors

Mathematics
for a Smarter Tomorrow

1 Definition

For two d -dimensional vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$, the Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}.$$

It represents the straight-line distance between the two points represented by $\mathbf{x}$ and $\mathbf{y}$ in d -dimensional space.

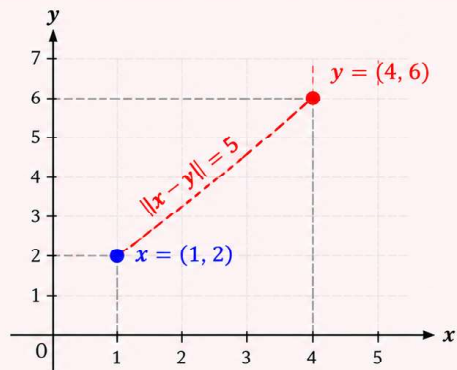
2 Numerical Example (2D)

Let $\mathbf{x} = (1, 2)$ and $\mathbf{y} = (4, 6)$. Then

$$\begin{aligned}\|\mathbf{x} - \mathbf{y}\| &= \sqrt{(4 - 1)^2 + (6 - 2)^2} \\ &= \sqrt{3^2 + 4^2} = \sqrt{25} = 5.\end{aligned}$$

So, the Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ is 5.

3 Geometric Interpretation (2D)



The Euclidean distance between the points $(1, 2)$ and $(4, 6)$ is the length of the line segment connecting them, which is 5.

The Euclidean distance measures how far apart two vectors (points) are in space.



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Mathematical Methods for Computing (CS-813)

Angle Between Two Vectors

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1 Definition

For two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^d$, the angle θ between them is defined using the dot product:

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

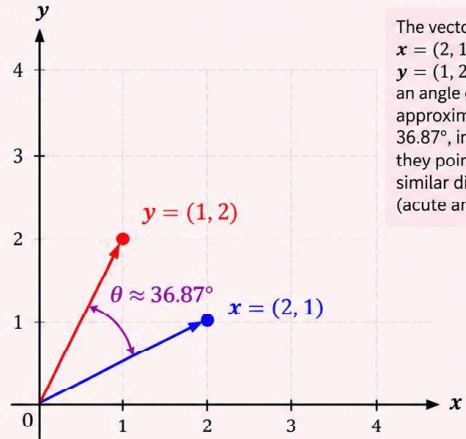
- $\theta \in [0, \pi]$ is the smaller angle between $\mathbf{x}$ and $\mathbf{y}$.
- If $\mathbf{x} \cdot \mathbf{y} > 0$, then $0 \leq \theta < 90^\circ$ (acute angle).
- If $\mathbf{x} \cdot \mathbf{y} = 0$, then $\theta = 90^\circ$ (orthogonal vectors).
- If $\mathbf{x} \cdot \mathbf{y} < 0$, then $90^\circ < \theta \leq 180^\circ$ (obtuse angle).

2 Numerical Example (2D)

Let $\mathbf{x} = (2, 1)$ and $\mathbf{y} = (1, 2)$. Find the angle between $\mathbf{x}$ and $\mathbf{y}$.

- Dot product: $\mathbf{x} \cdot \mathbf{y} = 2(1) + 1(2) = 4$
- Norms: $\|\mathbf{x}\| = \sqrt{2^2 + 1^2} = \sqrt{5}$, $\|\mathbf{y}\| = \sqrt{1^2 + 2^2} = \sqrt{5}$
- Cosine of the angle: $\cos \theta = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5}$
- Angle: $\theta = \cos^{-1}\left(\frac{4}{5}\right) \approx 36.87^\circ$

3 Geometric Interpretation (2D)



The vectors $\mathbf{x} = (2, 1)$ and $\mathbf{y} = (1, 2)$ make an angle of approximately 36.87° , indicating they point in similar directions (acute angle).



Orthogonal and Orthonormal Vectors

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for a Smarter Tomorrow

1 Orthogonal Vectors

Two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal (perpendicular) if their dot product is zero:

$$\mathbf{x} \cdot \mathbf{y} = 0$$

This implies that the angle between them is 90° .

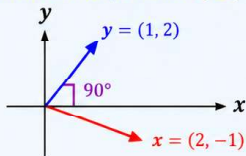
Example (2D)

Let $\mathbf{x} = (2, -1)$ and $\mathbf{y} = (1, 2)$. Then

$$\mathbf{x} \cdot \mathbf{y} = 2(1) + (-1)(2) = 2 - 2 = 0.$$

Hence, $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Geometric Interpretation (2D)



Since $\mathbf{x} \cdot \mathbf{y} = 0$,
the vectors are
perpendicular.

2 Orthonormal Vectors

Two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthonormal if they are orthogonal and each has unit length:

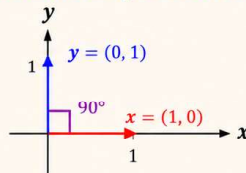
$$\mathbf{x} \cdot \mathbf{y} = 0 \quad \text{and} \quad \|\mathbf{x}\| = \|\mathbf{y}\| = 1$$

Example (2D)

Let $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (0, 1)$. Then

- $\mathbf{x} \cdot \mathbf{y} = 1(0) + 0(1) = 0$
- $\|\mathbf{x}\| = \sqrt{1^2 + 0^2} = 1, \quad \|\mathbf{y}\| = \sqrt{0^2 + 1^2} = 1$
- Hence, $\mathbf{x}$ and $\mathbf{y}$ are orthonormal.

Geometric Interpretation (2D)



Example in 3D

The standard basis vectors in $\mathbb{R}^3$

$$\mathbf{e}_1 = (1, 0, 0), \quad \mathbf{e}_2 = (0, 1, 0), \\ \mathbf{e}_3 = (0, 0, 1)$$

are mutually orthonormal, i.e.,

$$\mathbf{e}_i \cdot \mathbf{e}_j = 0 \quad \text{for } i \neq j, \quad \text{and} \quad \|\mathbf{e}_i\| = 1 \\ \text{for all } i = 1, 2, 3.$$



Examples in $\mathbb{R}^3$: Dot Product, Distance, Angle, Orthogonal and Orthonormal Vectors

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1 Vectors in $\mathbb{R}^3$

Let $\mathbf{a} = (1, 2, -1)$ and $\mathbf{b} = (2, -1, 2)$ in $\mathbb{R}^3$.

2 Dot Product

For $\mathbf{a} = (1, 2, -1)$ and $\mathbf{b} = (2, -1, 2)$,

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= 1(2) + 2(-1) + (-1)(2) \\ &= 2 - 2 - 2 = -2.\end{aligned}$$

Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, the vectors are not orthogonal.

3 Norms (Lengths)

$$\begin{aligned}\|\mathbf{a}\| &= \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6} \\ \|\mathbf{b}\| &= \sqrt{2^2 + (-1)^2 + 2^2} = 3\end{aligned}$$

4 Euclidean Distance

$$\begin{aligned}\|\mathbf{a} - \mathbf{b}\| &= \sqrt{(1-2)^2 + (2-(-1))^2 + (-1-2)^2} \\ &= \sqrt{(-1)^2 + 3^2 + (-3)^2} = \sqrt{19}.\end{aligned}$$

5 Angle Between Vectors

$$\begin{aligned}\cos \theta &= \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{-2}{\sqrt{6} \times 3} = \frac{-2}{3\sqrt{6}} \\ \theta &= \cos^{-1}\left(\frac{-2}{3\sqrt{6}}\right) \approx 106.26^\circ.\end{aligned}$$

The angle between $\mathbf{a}$ and $\mathbf{b}$ is approximately 106.26° , which is obtuse (greater than 90°).

6 Orthogonal Vectors (Example)

Let $\mathbf{u} = (1, 0, 1)$ and $\mathbf{v} = (1, 0, -1)$.

$$\mathbf{u} \cdot \mathbf{v} = 1(1) + 0(0) + 1(-1) = 0.$$

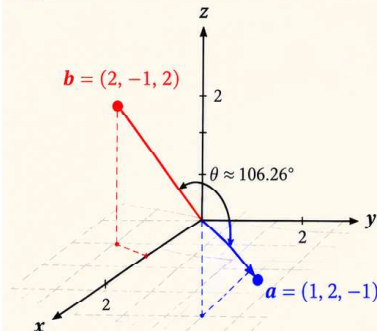
Hence, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors.

7 Orthonormal Vectors (Example)

Let $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$ and $\mathbf{e}_3 = (0, 0, 1)$.

- $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for $i \neq j$.
- $\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = \|\mathbf{e}_3\| = 1$.
- Thus, $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is an orthonormal set in $\mathbb{R}^3$ (the standard basis).

8 Geometric Interpretation in $\mathbb{R}^3$



Application: Angle and Cosine Similarity (Text Similarity in NLP)

Similarities
Connect Information

1 Problem Statement

In text analysis, we can represent a document as a vector of word counts (or weights). The angle between two document vectors measures their similarity.

Example: Consider two short documents with the vocabulary {data, science, machine}.

Document 1: "data science"

$$\mathbf{d}_1 = (2, 1, 0)$$

(data: 2, science: 1, machine: 0)

Document 2: "data machine"

$$\mathbf{d}_2 = (1, 0, 1)$$

(data: 1, science: 0, machine: 1)

Find the angle between the documents and their similarity using cosine similarity.



Real-World Use

In **search engines**, **recommendation systems**, and **text classification**, cosine similarity is used to find documents with similar content. A **smaller angle** (larger cosine similarity) means the documents are more similar.

2 Dot Product

The dot product measures how much the vectors point in the same direction.

$$\begin{aligned}\mathbf{d}_1 \cdot \mathbf{d}_2 &= 2(1) + 1(0) + 0(1) \\ &= 2 + 0 + 0 = 2\end{aligned}$$

3 Vector Norms (Lengths)

Compute the Euclidean norms of the vectors.

$$\begin{aligned}\|\mathbf{d}_1\| &= \sqrt{2^2 + 1^2 + 0^2} = \sqrt{5} \\ \|\mathbf{d}_2\| &= \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}\end{aligned}$$

4 Angle Between the Vectors

Use the cosine formula:

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|} = \frac{2}{\sqrt{5} \cdot \sqrt{2}} = \frac{2}{\sqrt{10}}$$

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{10}}\right) \approx \cos^{-1}(0.6325) \approx 50.8^\circ$$

5 Cosine Similarity

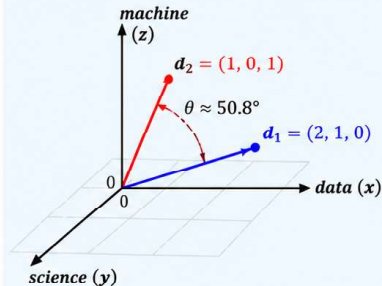
Cosine similarity is the cosine of the angle:

$$\text{similarity}(\mathbf{d}_1, \mathbf{d}_2) = \cos \theta = \frac{2}{\sqrt{10}} \approx 0.6325$$

The similarity value lies between -1 and 1 .

Here, 0.6325 indicates a moderate similarity between the two documents.

6 Geometric Interpretation (3D)



Application of Euclidean Distance: Nearest Neighbour Classification

Mathematics
for a Smarter Tomorrow

1 Problem Statement

We have labelled data points belonging to two classes and a new point. Use Euclidean distance to find the nearest neighbour of the new point and predict its class.

2 Data Points (2D)

Consider the following points in $\mathbb{R}^2$:

Point	Coordinates (x, y)	Class
A	(1, 2)	Class 1
B	(2, 1)	Class 1
C	(4, 4)	Class 2
D	(5, 3)	Class 2
P (new)	(3, 2)	?

3 Euclidean Distance Formula

For two points $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$,

$$d(x, y) = \|x - y\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

The Euclidean distance measures the straight-line distance between two points in space.

4 Step-by-Step Calculation

Compute the distance from $P(3, 2)$ to each labelled point.

Distance to A(1, 2):

$$\begin{aligned}d(P, A) &= \sqrt{(3-1)^2 + (2-2)^2} \\&= \sqrt{4+0} = 2\end{aligned}$$

Distance to B(2, 1):

$$\begin{aligned}d(P, B) &= \sqrt{(3-2)^2 + (2-1)^2} \\&= \sqrt{1+1} = \sqrt{2} \approx 1.414\end{aligned}$$

Distance to C(4, 4):

$$\begin{aligned}d(P, C) &= \sqrt{(3-4)^2 + (2-4)^2} \\&= \sqrt{1+4} = \sqrt{5} \approx 2.236\end{aligned}$$

Distance to D(5, 3):

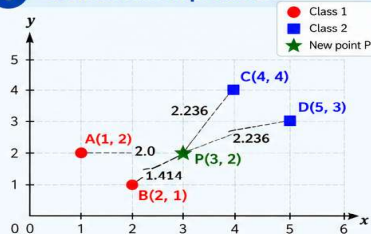
$$\begin{aligned}d(P, D) &= \sqrt{(3-5)^2 + (2-3)^2} \\&= \sqrt{4+1} = \sqrt{5} \approx 2.236\end{aligned}$$

5 Compare Distances

The distances from P to all points are:

Point	Distance	Class
A	2.000	Class 1
B	1.414	Class 1
C	2.236	Class 2
D	2.236	Class 2

6 Geometric Interpretation



7 Result and Application

The nearest neighbour of $P(3, 2)$ is point $B(2, 1)$, with distance $\sqrt{2} \approx 1.414$.

$P(3, 2) \rightarrow$ Class 1

Applications



Text classification

Identify the most similar document in a collection.



Recommendation systems

Find similar users or items.



Image retrieval

Retrieve similar images using feature vectors.



Healthcare

Find similar patient records for diagnosis or treatment.



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Mathematical Methods for Computing (CS-813)

Manhattan Distance (L1 Distance)

Mathematics
for a Smarter Tomorrow

1 Definition

For two points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ in $\mathbb{R}^n$, the Manhattan distance (L_1 distance) is

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^n |x_i - y_i|.$$

It is called Manhattan distance because it measures the distance along grid-like paths, like the streets in a city.

4 Numerical Example (3D)

Let $P = (2, -1, 3)$ and $Q = (5, 4, 1)$.

The Manhattan distance is

$$\begin{aligned} d_1(P, Q) &= |5 - 2| + |4 - (-1)| + |1 - 3| \\ &= 3 + 5 + 2 \\ &= 10. \end{aligned}$$

2 Numerical Example (2D)

Let $A = (1, 2)$ and $B = (4, 6)$.

The Manhattan distance is

$$\begin{aligned} d_1(A, B) &= |4 - 1| + |6 - 2| \\ &= 3 + 4 \\ &= 7. \end{aligned}$$

5 Comparison with Euclidean Distance

For the points $A = (1, 2)$ and $B = (4, 6)$:

Manhattan distance:

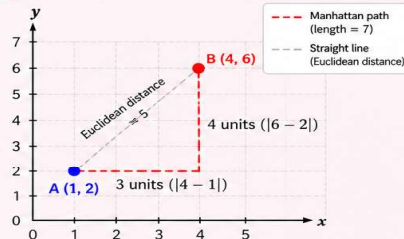
$$d_1(A, B) = 7.$$

Euclidean distance:

$$\begin{aligned} d_2(A, B) &= \sqrt{(4 - 1)^2 + (6 - 2)^2} \\ &= \sqrt{3^2 + 4^2} = 5. \end{aligned}$$

Here, the Manhattan distance is larger than the Euclidean distance because it restricts movement to horizontal and vertical directions.

3 Geometric Interpretation (2D)



6 Applications



Path planning and navigation

Used in grid-based maps (e.g., city streets, robotics, game AI).



Machine learning

Used as a distance metric in clustering (e.g., k-NN) and high-dimensional data.



Text analysis

Measures differences between word count vectors (e.g., in information retrieval).



Image processing

Used to compare images represented as pixel intensity vectors.



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Mathematical Methods for Computing (CS-813)

Application of Manhattan Distance in Machine Learning

Example: k -Nearest Neighbour (k -NN) Classification

Mathematics
for a Smarter Tomorrow

1 Problem Statement

We have labelled data points from two classes and a new point. Using Manhattan distance (L_1), find the nearest neighbour of the new point and predict its class using 1-NN.

2 Dataset (2D)

Each point is represented by two features (x_1, x_2).

Point	Coordinates (x_1, x_2)	Class
A	(1, 2)	Class 0
B	(4, 1)	Class 0
C	(3, 5)	Class 1
D	(6, 4)	Class 1
Q (new)	(5, 2)	?

3 Manhattan Distance Formula

For two points $x = (x_1, x_2)$ and $y = (y_1, y_2)$,

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

It measures the distance along horizontal and vertical directions (grid-like paths).

4 Step-by-Step Calculation

Compute the Manhattan distance from $Q(5, 2)$ to each labelled point.

Distance to A(1, 2):

$$d_1(Q, A) = |5 - 1| + |2 - 2| \\ = 4 + 0 = 4$$

Distance to B(4, 1):

$$d_1(Q, B) = |5 - 4| + |2 - 1| \\ = 1 + 1 = 2$$

Distance to C(3, 5):

$$d_1(Q, C) = |5 - 3| + |2 - 5| \\ = 2 + 3 = 5$$

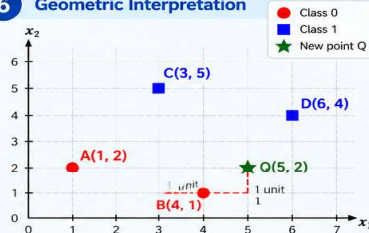
Distance to D(6, 4):

$$d_1(Q, D) = |5 - 6| + |2 - 4| \\ = 1 + 2 = 3$$

5 Distances and Nearest Neighbour

Point	Manhattan Distance to Q(5, 2)	Class
A	4	Class 0
B	2	Class 0
C	5	Class 1
D	3	Class 1

6 Geometric Interpretation



7 Result and Application

The nearest neighbour of $Q(5, 2)$ is point $B(4, 1)$ with Manhattan distance 2.

$Q(5, 2) \rightarrow$ Class 0

8 Applications in Machine Learning



k-NN classification

Predict the class of a new data point using nearest neighbours.



Text classification

Measure distance between word count vectors using Manhattan distance.



Recommender systems

Compare user/item feature vectors (e.g., ratings).



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